

Problem set #5

MATH 4503/6503: Numerical Methods for Differential Equations

Due date: April 12, 2026

Reading resources: Notes uploaded to D2L; book pages 13-32 and chapters 9, 10, and 11

Problem #1: von Neumann (the Great) analysis

Hello, friends! Welcome to the last of your problem sets. You must feel quite relieved I know! For me though, this farewell is something of a sad one as you have been an exceptional class, extremely fun to teach.

Anyway, let us get into the problems and get them done with. This first problem is on the von Neumann stability analysis. Consider the smallest possible stencil for the diffusion equation that is centered both in space and in time. Using the von Neumann analysis, show that such a stencil is unconditionally unstable.

Note: I have tried in class to impress upon you the remarkable stature John von Neumann holds in both physics and mathematics (and more), but I do not think such introductions can ever do him justice. Therefore, I invite you to read Ananyo Bhattacharya's beautiful biography of his: *The man from the future*.

Problem #2: tridiagonal systems

This problem is an exercise in using Maple to set up matrix-vector equations in numerical methods for PDEs and solving them. Consider the n -dimensional vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{d}$:

$$\mathbf{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}, \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}, \mathbf{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}, \mathbf{d} = \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix}.$$

These can be used to define a linear system of equations for $\mathbf{x}^T = [x_1, \dots, x_n]$:

$$\begin{bmatrix} a_1 & c_1 & & & \\ b_2 & a_2 & c_2 & & \\ & b_3 & a_3 & c_3 & \\ & & \ddots & \ddots & \ddots \\ & & & b_{n-1} & a_{n-1} & c_{n-1} \\ & & & & b_n & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{n-1} \\ d_n \end{bmatrix}.$$

Notice that the coefficient matrix on the left is tridiagonal (three nonzero diagonals and zero everywhere else); as we have seen in class, systems of this type arise often in the numeric solution of PDEs. Also notice how b_1 and c_n do not actually enter the system.

- (a) Write a procedure that takes **a**, **b**, **c**, and **d** as arguments and returns the solution **x** of the above linear system. The procedure should make no use of high level Maple commands, such as `solve`, `isolate`, `fsolve`, `MatrixInverse`, `LinearSolve`, or anything in the `LinearAlgebra` package for solution purposes.

Hint: Look up Thomas algorithm in case you do not remember your linear algebra Gaussian elimination well.

- (b) Compare the results of your procedure to the output of `LinearSolve` for the case $a_i = -2$, $b_i = c_i = 1$, where $i = 1, \dots, n$, and with **d** a random vector; that is,

$$\mathbf{d} = \text{LinearAlgebra}[\text{RandomMatrix}](1, n).$$

(Obviously, you need to make a specific choice of n for this to work.) Feel free to use `BandMatrix` to generate the coefficient matrix that is the first argument of `LinearSolve`.

Problem #3: a 1D elliptic problem

In quantum mechanics, the energy E of a particle trapped in a 1D potential $V(x)$ with infinite walls at $x = \pm 1$ is a solution of an ODE eigenvalue problem:

$$Eu(x) = -u''(x) + V(x)u(x), u(-1) = u(1) = 0.$$

All possible energies then form a set of numbers that the above 1D boundary value problem (that is, a 1D version of an elliptic PDE) has a solution for.

- (a) By introducing a lattice for $x \in [-1, 1]$ with N interior nodes and employing a centered finite difference approximation for $u''(x)$, convert the above ODE eigenvalue problem into an N -dimensional matrix eigenvalue problem.
- (b) Write a Maple procedure that solves for the energies (each denoted by E above) for an arbitrary potential $V(x)$ and number of nodes.

Hint: You might want to use the Maple command `LinearAlgebra[Eigenvalues]`.

- (c) Let us look at a truncated simple harmonic oscillator with $V(x) = V_0 x^2$ for $x \in [-1, 1]$. Plot the lowest ten approximations for the energy eigenvalues as a function of N when $V_0 = 50$.